

5(a)(i)	correct circuit symbol for a diode shown correctly connected in series with the wires leading into and out of the dotted box	B1
5(a)(ii)	smoothing / V_{OUT} is smoothed	B1
5(b)(i)	frequency = $1 / 0.04$ = 25 Hz	A1
5(b)(ii)	$V = V_0 \exp(-t / RC)$ and $\tau = RC$ or $V = V_0 \exp(-t / \tau)$	C1
	$3.25 = 5.50 \exp(-0.020 / \tau)$ leading to $\tau = 0.038$ s	A1
5(b)(iii)	$\tau = RC$	C1
	capacitance = $0.038 / 14000$ = 2.7×10^{-6} F	A1
5(c)	V_{IN} has constant magnitude in both positive and negative directions	B1
	(so) V_{OUT} is (now) constant / V_{OUT} does not vary with time	B1